

# Task 85: Combinations

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## PROBLEM DESCRIPTION

Given two integers  $n$  and  $k$ , return all possible combinations of  $k$  numbers chosen from the range  $[1, n]$ . You may return the answer in any order.

## JAVA SOLUTION

```
import java.util.ArrayList;
import java.util.List;

class Solution {
    public List<List<Integer>> combine(int n, int k) {
        List<List<Integer>> result = new ArrayList<>();
        backtrack(result, new ArrayList<>(), 1, n, k);
        return result;
    }

    private void backtrack(List<List<Integer>> result, List<Integer>
tempList, int start, int n, int k) {
        if (tempList.size() == k) {
            result.add(new ArrayList<>(tempList));
            return;
        }

        for (int i = start; i <= n; i++) {
            tempList.add(i);
            backtrack(result, tempList, i + 1, n, k);
            tempList.remove(tempList.size() - 1);
        }
    }
}
```